

Gated Adaptive Hierarchical Attention Unit Neural Networks for the Life Prediction of Servo Motors

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Abstract—The remaining useful life (RUL) prediction of servo motors plays a greatly indispensable role in preventing abnormal and inefficient operation of servo drive system and even major safety accident. In this article, a novel variant of recurrent neural networks, called gated adaptive hierarchical attention unit (GAHAU), is constructed to efficiently predict the RULs of servo motors, which can not only pay attention to the outputs of reset and update gates but also adaptively divide the hierarchy of the sequence information in attention gate. The updating formulas of GAHAU are derived. With the raw vibration signals of servo motors, a quadratic function-based deep convolutional autoencoder is used for automatically generating the health indicator vector. Based on the constructed HI vector, the RULs of servo motors are accurately predicted by GAHAU. The experiments on servo motors demonstrate that the presented GAHAU has remarkable prediction performance especially for the long-term prediction. Compared to the existing typical RUL prediction methods, the proposed methodology has stronger predictive ability according to the comparative results.

Index Terms—Attention hierarchy mechanism, health indicator (HI), long-term prediction, remaining useful life (RUL) prediction, servo motor.

I. INTRODUCTION

AS THE most crucial part of the precision equipment, servo motor has been widely applied in varied fields, such as numerical control machine tools, industrial robots [1]. Since the complicated and uncontrollable working environment, the servo motor is tremendously prone to performance degradation or fault, and the servo drive equipment will work abnormally in such case. The performance degradation of servo motor always leads to quality decline and productivity reduction. When the

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serious fault occurs, it even results in catastrophe [2]. Consequently, the remaining useful life (RUL) prediction of servo motor has significant engineering value [3].

As the core of the prognostics and health management, the RUL prediction approaches can be coarsely classified into two main categories: Model-based and data-driven approaches [4], [5]. The model-based approaches aim to construct a physical model to actually reflect the degradation trend of machine when the fault mechanism is fully understood via a simulation experiment. However, the physical model may not be fit for the complex and harsh working condition. Data-driven methods attempt to utilize the statistical and machine learning technologies to mine the regression relationship between the degradation performance and the monitoring data. In addition, data-driven approaches do not need the expert knowledge, so they can be broadly employed for predicting the RULs of complicated systems, whose fault mechanism is difficultly revealed. For instance, Huang *et al.* [6] built a self-organizing map network to automatically generate a novel degradation indicator named minimum quantization error (MQE), then an RUL prediction model based on MQE and back propagation (BP) network was developed for the bearing RUL prediction. Loutas *et al.* [7] proposed a support vector regression model for the RUL prediction of rolling bearings, which used the statistical features from multiple domains as the input. Chen *et al.* [8] explored a novel prediction approach that combined the adaptive neuro-fuzzy inference systems and high-order particle filtering.

Recently, deep learning (DL) technology has already attracted substantial attention in the field of RUL prediction. Many DL-based methods have been developed for RUL prediction. This is mainly because that the DL model has strong fitting ability to effectively build the potential relationship between RUL and measured data. Zhu *et al.* [9] developed a multiscale convolution neural network for bearing RUL prediction, and it could learn global and local feature information synchronously. Wang *et al.* [10] put forward a deep separable convolutional network which used the data acquired from different sensors as input to predict the machinery RUL. Yang *et al.* [11] constructed a double-convolutional neural network for RUL prediction. The purpose of the first convolutional neural network (CNN) is to identify the incipient fault point of bearing, and the next CNN model is directly applied to predict the RUL. Chen *et al.* [12] proposed an attention-based DL method that automatically learnt health features from the raw sensory data, and then the

RULs of aircraft engines were estimated by the extracted and handcrafted features. Zhao *et al.* [13] presented a probabilistic RUL prediction approach by deep convolutional neural network, and it could precisely output the probability density of the actual RUL instead of a single RUL value. Zhu *et al.* [14] used the hidden Markov model to determine the location of the fault occurrence time, then a transfer learning model with multiple layer perceptron was constructed to predict RULs under different working conditions. Liu and Gryllias [15] combined the domain adversarial neural network (DANN) with the bidirectional long short-term memory (Bi-LSTM) network to forecast the RULs of bearings.

In practice, RUL prediction can be regarded as a problem of time series prediction. Several types of recurrent neural networks (RNNs), such as gated recurrent unit (GRU) [16] and LSTM [17], can be used to address this issue. Owing to the particular gate mechanism, both GRU and LSTM possess the unique ability of processing long-term information. Whereupon, they have been successfully applied in various industrial fields, including gear [18], bearing [19], turbofan engine [20], wind turbine [21], and so on. Qin *et al.* [18] proposed a macro-micro attention mechanism and designed the macro-micro attention LSTM (MMALSTM) network to have differential learning capability for significant information. Ma and Mao [22] proposed a convolution-based LSTM (CLSTM) network that had the ability to capture long-term information and synchronously extract features from time-frequency domain. Ren *et al.* [23] presented an end-to-end approach based on multiscale dense GRU (MDGRU) for bearing RUL prediction.

Attention mechanism, as a resource allocation scheme, can filter information by the use of the limited computing source, and address the problem of information overload. A variety of attention-based RNNs have been explored for RUL prediction. A time series multiple channel convolutional neural network with attention-based LSTM (TSMC-CNN-ALSTM) was built for bearing RUL prediction [24], and this network could pay attention to different extracted features at different time steps. A bidirectional GRU (Bi-GRU) with attention mechanism was proposed to generate the health indicator (HI) values, then they were used as the input of linear regression for RUL prediction [25]. An end-to-end approach based on feature-attention, Bi-GRU and CNN was put forward to predict the machinery RUL [26]. A novel RUL prediction network named gated dual attention unit (GDAU) was constructed for predicting the bearing RUL, and it could simultaneously pour attention into significant information of input data, historical data, the outputs of reset and update gates [27].

Owing to the different capacities of various gates in GRU, it can be directly used for long-term RUL prediction. But it does not have the ability of dealing with crucial information, especially when different information need to be synchronously processed in the training of GRU network. Aiming at the abovementioned problem, the attention mechanism provides a feasible way. As mentioned before, several attention-based RUL prediction methods have been explored in the past few years. However, these RUL prediction methods do not consider the sequence information of the neural network, that is, the sequence

of neurons is unordered. For example, the output of reset gate in GRU is a vector, whereas all elements are disorderly placed in this vector. It is easy to understand that the sequence of neurons is related to time-varying information (e.g., health status), especially in the time-series prediction networks. Therefore, it is a feasible way to enhance the predictive ability by taking full advantage of the sequence information. This work develops a novel variant of GRU named as gated adaptive hierarchical attention unit (GAHAU) to predict the RULs of servo motors. It can not only extract the attention information from the outputs of the reset and update gates but also adaptively divide the sequence information of attention gates into three hierarchies to update the attention distribution. In these attention hierarchies, the high attention hierarchy represents the more significant attention information (i.e., long-term tendency) that should be kept in the attention gate; the middle attention hierarchy represents a normal variation trend (i.e., long-short-term tendency) that needs to be renewed by a specific updating mechanism; and the low attention hierarchy represents the less significant attention information (i.e., short-term tendency) that needs to be also updated in the attention gate. Owing to the above attention hierarchies, GAHAU possesses a powerful prediction capacity.

Using the acquired vibration signals of servo motor, the fast Fourier transform (FFT) datasets of raw signals are calculated. With these FFT datasets, the HIs of servo motors are constructed via a quadratic function-based deep convolutional autoencoder (QFDCAE) network [28]. After that, a GAHAU-based RUL prediction method for servo motors is developed in this study. The comparative results prove that the proposed methodology is superior to other typical RUL prediction approaches, especially for long-term RUL prediction. The main contributions of this work are summarized as below.

- 1) An adaptive hierarchical attention unit is first proposed to automatically divide the sequence information of attention gate into high attention, middle attention, and low attention hierarchies.
- 2) With the adaptive attention hierarchy mechanism, GAHAU neural network is designed, and the updating formulas are derived. GAHAU can not only pay attention to the trend information embedded into the reset and update gates but also adaptively mine the sequence information of attention gate, so it possesses both long-term and short-term predictive ability.
- 3) Combining the servo motor HI constructed by the QFDCAE network with the GAHAU network, a novel RUL prediction methodology for servo motors is presented, which has higher prediction performance than other typical methods.

II. RELATED THEORIES

In this section, the theoretical background of gated attention unit (GAU) [27] is introduced for the construction of GAHAU framework.

As one of the GRU's variants, GAU can pay attention to the significant information of the outputs of the reset and update

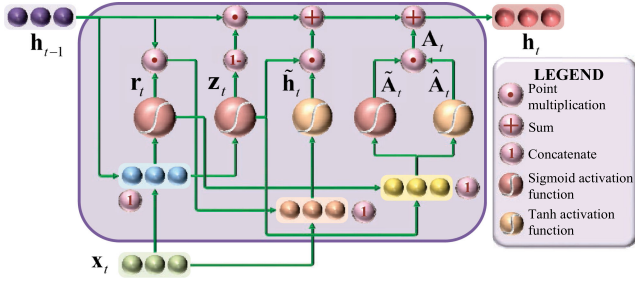


Fig. 1. Cell architecture of GAU network.

gates due to its attention gate. The cell structure of GAU network is displayed in Fig. 1.

From Fig. 1, we can note that the temporary hidden state, the reset, and update gates of GRU are retained in GAU network, but the hidden state $\mathbf{h}_t$ in the cell of GAU is reformulated as

$$\begin{aligned}
 \mathbf{r}_t &= \sigma(\mathbf{W}_{rx}\mathbf{x}_t + \mathbf{U}_{rh}\mathbf{h}_{t-1} + \mathbf{b}_r) \\
 \mathbf{z}_t &= \sigma(\mathbf{W}_{zx}\mathbf{x}_t + \mathbf{U}_{zh}\mathbf{h}_{t-1} + \mathbf{b}_z) \\
 \tilde{\mathbf{h}}_t &= \phi(\mathbf{W}_{hx}\mathbf{x}_t + \mathbf{U}_{hh}(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \\
 \tilde{\mathbf{A}}_t &= \sigma(\mathbf{W}_{\tilde{A}r}\mathbf{r}_t + \mathbf{U}_{\tilde{A}z}\mathbf{z}_t + \mathbf{b}_{\tilde{A}}) \\
 \hat{\mathbf{A}}_t &= \phi(\mathbf{W}_{\hat{A}r}\mathbf{r}_t + \mathbf{U}_{\hat{A}z}\mathbf{z}_t + \mathbf{b}_{\hat{A}}) \\
 \mathbf{A}_t &= \tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t \\
 \mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + (\mathbf{z}_t \odot \tilde{\mathbf{h}}_t) + \mathbf{A}_t
 \end{aligned} \quad (1)$$

where $\sigma(x)$ and $\phi(x)$ are, respectively, the logistic sigmoid and hyperbolic tangent function; $\mathbf{x}_t$ is the input data at the time instant t ; $\mathbf{h}_{t-1}$ denotes the hidden state at the time instant $t-1$; $\mathbf{r}_t$ and $\mathbf{z}_t$ are the outputs of reset and update gates, respectively; $\mathbf{W}$ and $\mathbf{U}$ denote weight matrixes (e.g., $\mathbf{W}_{rx}$ and $\mathbf{U}_{rh}$, respectively, represent the matrixes of weights relating to the input data $\mathbf{x}_t$ and the hidden state $\mathbf{h}_{t-1}$); $\mathbf{b}$ denotes the bias matrix (e.g., $\mathbf{b}_r$ is the bias matrix associating with reset gate); $\tilde{\mathbf{h}}_t$ denotes the temporary hidden state at time instant t ; $\tilde{\mathbf{A}}_t$ denotes the ratio of attention on the outputs of reset and update gates; $\hat{\mathbf{A}}_t$ determines whether the outputs of the two gates are positive or negative; $\mathbf{A}_t$ represents the attention distribution on these two gates; and $\odot$ represents dot product.

III. PROPOSED METHODOLOGY

For the proposed methodology, the servo motor HI vector is first constructed by the QFDCAE network; then GAHAU is developed for the prediction of the HI time series based on the obtained HI vector; finally the detailed procedure of servo motor RUL prediction is given.

A. Servo Motor Health Indicator Construction

The vibration signals of servo motors contain a mass of status information, which can be easy to collect. Compared to electrical signals, vibration signals can better reflect the overall performance of servo motors. And the motor vibration can also

be easy to collect. Therefore, this study uses vibration signals for the motor RUL prediction. To obtain a good predictive result, HI construction plays an important role in the RUL prediction. Via comprehensively comparing the varied HI construction methods, we find that the supervised QFDCAE neural networks is more suitable for constructing the servo motor HI, and the obtained HI has more obvious degradation trend. It then follows that QFDCAE based on vibration data is utilized to generate the motor HI. There are 27 layers in the QFDCAE network, including an input layer, an output layer, 13 convolution layers, six pooling layers, and six upsampling layers. In the encoder of QFDCAE, the numbers of convolution kernels are, respectively, 20, 40, 80, 40, 20, 10, and 1. On the contrary, the numbers of convolutional kernels are, respectively, 10, 20, 40, 80, 40, and 20 in the decoder of QFDCAE. The detail of this method can be seen in [28].

B. Gated Adaptive Hierarchical Attention Unit

As previously mentioned, GAU can pay attention to the outputs of the reset and update gates, whereas the sequence information of the attention gate in GAU is not considered. In order to adaptively process the sequence information, we construct a novel neural network, namely, GAHAU, which can divide the sequence information of attention gate into high attention hierarchy, middle attention hierarchy, and low attention hierarchy. The cell structure of GAHAU is illustrated in Fig. 2.

Inspired by the ordered neurons LSTM (ON-LSTM) network [29], we design a hierarchical mechanism for GAU. Via the outputs of the reset and update gates, the hierarchical position d_1 of the high attention hierarchy information and the hierarchical position d_2 of the low attention hierarchy information are separately calculated as

$$d_1 = f_1(\mathbf{r}_t, \mathbf{z}_t) \quad (2)$$

$$d_2 = f_2(\mathbf{r}_t, \mathbf{z}_t) \quad (3)$$

where f_1 and f_2 are two predictors for estimating the hierarchical position d_1 and d_2 , respectively. And the corresponding realization process will be presented in the following. With the two hierarchical positions, high attention, middle attention, and low attention hierarchies are determined, then the attention distribution $\mathbf{A}_t$ on the reset and update gates can be updated. Considering the relative magnitudes between d_1 and d_2 , different updating formulations can be derived.

The high attention hierarchy is defined by d_1 , while the low attention hierarchy is determined by d_2 . If $d_1 \leq d_2$, there is an overlapping region $[d_1, d_2]$ between high and low attention hierarchies, that is, the middle attention hierarchy. Fig. 3 illustrates the hierarchical division of the sequence information of attention gate in such case. From (1), $\hat{\mathbf{A}}_t$ represents the positive or negative information due to the value range $(-1, 1)$ of Tanh activation function. As the short-term trend may have positive or negative effect on the trend prediction, $\hat{\mathbf{A}}_t$ can be regarded as the short-term information. Similarly, $\tilde{\mathbf{A}}_t$ just contains the positive information with small fluctuation owing to the value range $(0, 1)$ of Sigmoid activation function, thus it mainly represents the long-term tendency which changes slowly. Therefore, $\hat{\mathbf{A}}_t$

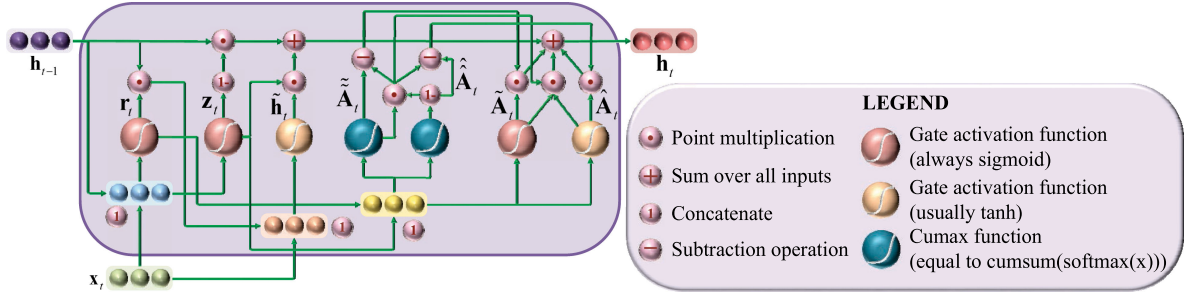
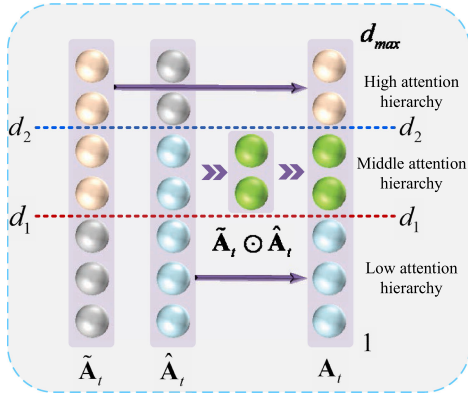
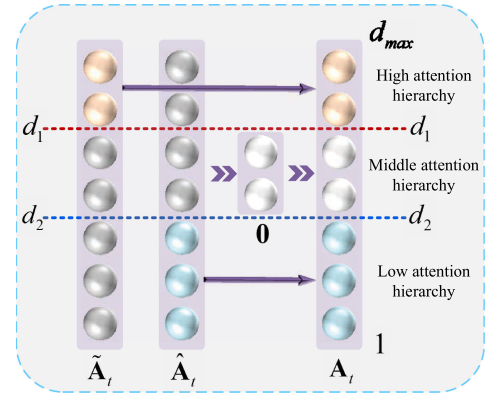


Fig. 2. Cell structure of GAHAU network.

Fig. 3. Hierarchical division of attention gate when $d_1 \leq d_2$.Fig. 4. Hierarchical division of attention gate when $d_1 \geq d_2$.

is used to update the attention distribution in the low-attention hierarchical region of $[0, d_1]$, while $\tilde{\mathbf{A}}_t$ is utilized to update the attention distribution in the high-attention hierarchical region of $(d_2, d_{\max}]$, where $d_{\max}$ denotes the dimension of the attention distribution. In the overlapping region of $[d_1, d_2]$, the attention distribution is updated by $\mathbf{A}_t$ shown in (1). In summary, when $d_1 \leq d_2$, the attention distribution $\mathbf{A}_t$ based on attention hierarchy is updated by

$$\mathbf{A}_t = \begin{pmatrix} \tilde{\mathbf{A}}_t, (d_2, d_{\max}] \\ \tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t, [d_1, d_2] \\ \hat{\mathbf{A}}_t, [0, d_1] \end{pmatrix}. \quad (4)$$

If $d_1 \geq d_2$, there is no overlapping region between the low and high attention hierarchies. In such case, the hierarchical division is demonstrated in Fig. 4. In the low-attention hierarchical region $[0, d_2]$, the attention distribution is still updated by $\hat{\mathbf{A}}_t$; and in the high-attention hierarchical region $(d_1, d_{\max}]$, it is still updated by $\tilde{\mathbf{A}}_t$. However, in the middle-attention hierarchical region $[d_2, d_1]$, the attention distribution $\mathbf{A}_t$ is directly set to $\mathbf{0}$, since there is no overlapping region (i.e., long-short-term information) in this case. To sum up, when $d_1 \geq d_2$, the attention distribution $\mathbf{A}_t$ is calculated by

$$\mathbf{A}_t = \begin{pmatrix} \tilde{\mathbf{A}}_t, (d_1, d_{\max}] \\ \mathbf{0}, [d_2, d_1] \\ \hat{\mathbf{A}}_t, [0, d_2] \end{pmatrix}. \quad (5)$$

In order to implement the adaptive update of the aforementioned attention distribution, we first introduce two vectors through the outputs of the reset and update gates, which are, respectively, defined as

$$\mathbf{A}_{d_1} = \text{softmax}(\mathbf{W}_{\tilde{\mathbf{A}}r} \mathbf{r}_t + \mathbf{U}_{\tilde{\mathbf{A}}z} \mathbf{z}_t + \mathbf{b}_{\tilde{\mathbf{A}}}) \quad (6)$$

$$\mathbf{A}_{d_2} = \text{softmax}(\mathbf{W}_{\hat{\mathbf{A}}r} \mathbf{r}_t + \mathbf{U}_{\hat{\mathbf{A}}z} \mathbf{z}_t + \mathbf{b}_{\hat{\mathbf{A}}}). \quad (7)$$

Suppose that $\mathbf{x} = [x_1, x_2, x_3, \dots, x_n]$, a cumsum function is defined as $\text{cumsum}(\mathbf{x}) = [x_1, x_1 + x_2, \dots, x_1 + \dots + x_n]$. Then, to make sure that the values of $\tilde{\mathbf{A}}_t$ are monotonically increasing from 0 to 1, while the values of $\hat{\mathbf{A}}_t$ are monotonically decreasing from 1 to 0, the master attention gates $\tilde{\mathbf{A}}_t$ and $\hat{\mathbf{A}}_t$ are, respectively, formulated as

$$\tilde{\mathbf{A}}_t = \text{cumsum}(\mathbf{A}_{d_1}) \quad (8)$$

$$\hat{\mathbf{A}}_t = 1 - \text{cumsum}(\mathbf{A}_{d_2}). \quad (9)$$

Via (8) and (9), three different hierarchies can be divided via the following equations:

$$\xi_{t1} = \tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t \quad (10)$$

$$\xi_{t2} = \tilde{\mathbf{A}}_t - \xi_{t1} \quad (11)$$

$$\xi_{t3} = \hat{\mathbf{A}}_t - \xi_{t1} \quad (12)$$

where ξ_{t1} , ξ_{t2} , and ξ_{t3} denote the middle attention hierarchy, the high attention hierarchy, and low attention hierarchy, respectively.

By using (1) and (6)–(12), the hidden state $\mathbf{h}_t$ of GAHAU at the time instant t is obtained as

$$\begin{aligned} \mathbf{r}_t &= \sigma(\mathbf{W}_{rx}\mathbf{x}_t + \mathbf{U}_{rh}\mathbf{h}_{t-1} + \mathbf{b}_r) \\ \mathbf{z}_t &= \sigma(\mathbf{W}_{zx}\mathbf{x}_t + \mathbf{U}_{zh}\mathbf{h}_{t-1} + \mathbf{b}_z) \\ \tilde{\mathbf{h}}_t &= \phi(\mathbf{W}_{hx}\mathbf{x}_t + \mathbf{U}_{hh}(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \\ \tilde{\mathbf{A}}_t &= \sigma(\mathbf{W}_{\tilde{A}r}\mathbf{r}_t + \mathbf{U}_{\tilde{A}z}\mathbf{z}_t + \mathbf{b}_{\tilde{A}}) \\ \hat{\mathbf{A}}_t &= \phi(\mathbf{W}_{\hat{A}r}\mathbf{r}_t + \mathbf{U}_{\hat{A}z}\mathbf{z}_t + \mathbf{b}_{\hat{A}}) \\ \mathbf{A}_t &= \xi_{t3} \odot \hat{\mathbf{A}}_t + \xi_{t1} \odot (\tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t) + \xi_{t2} \odot \tilde{\mathbf{A}}_t \\ \mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + (\mathbf{z}_t \odot \tilde{\mathbf{h}}_t) + \mathbf{A}_t. \end{aligned} \quad (13)$$

Then the output of GAHAU can be calculated by

$$\mathbf{y}_t = H(\mathbf{W}\mathbf{h}_t + \mathbf{b}) \quad (14)$$

where $\mathbf{y}_t$ is the output of the GAHAU; H is the linear activation function; $\mathbf{W}$ and $\mathbf{b}$ are, respectively, the weight matrix and bias vector.

According to the output vector $\mathbf{y}_t$, the loss function of the proposed GAHAU network is defined as

$$\mathcal{L}(\mathbf{y}_t, \hat{\mathbf{y}}_t) = \frac{1}{N} \sum_{n=1}^N \left(y_t^{(n)} - \hat{y}_t^{(n)} \right)^2 \quad (15)$$

where $\hat{\mathbf{y}}_t$ is the actual HI vector and N is the length of the output vector.

In order to explain the principle of this new attention distribution update rule, we suppose that the master gates are binary vectors. Suppose that the master attention gate $\tilde{\mathbf{A}}_t = (0, \dots, 0, 1, \dots, 1)$ is a $d_{\max}$ -dimension vector and the hierarchical position of the high attention hierarchy information is d_1 . We can note that the elements before the index d_1 are 0, while the elements from d_1 to $d_{\max}$ are 1 in $\tilde{\mathbf{A}}_t$. Similarly, suppose that the master attention gate $\hat{\mathbf{A}}_t = (1, \dots, 1, 0, \dots, 0)$ is also a $d_{\max}$ -dimension vector and the hierarchical position of the low attention hierarchy information is d_2 . The elements of $\hat{\mathbf{A}}_t$ are defined as 0 when the index is greater than d_2 . In the proposed method, the master attention gate $\tilde{\mathbf{A}}_t$ denotes the high attention information, while the master attention gate $\hat{\mathbf{A}}_t$ denotes the low attention information. The element of 0 means paying no attention to information, while the element of 1 means paying attention to information. $\xi_{t1} = \tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t$ denotes the overlap of $\tilde{\mathbf{A}}_t$ and $\hat{\mathbf{A}}_t$, and the elements of ξ_{t1} in the overlap region are taken as 1 while the other elements are equal to 0. Especially, it is a zero vector when there is no overlap. First, since $\tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t$ denotes the long-short-term tendency, $\xi_{t1} \odot (\tilde{\mathbf{A}}_t \odot \hat{\mathbf{A}}_t)$ is employed to update the attention distribution $\mathbf{A}_t$ within the overlap region, i.e., the middle attention hierarchy. Second, $\xi_{t2} = \tilde{\mathbf{A}}_t - \xi_{t1}$ represents a vector whose elements in

the range of $(\max(d_1, d_2), d_{\max}]$ are 1, and the rest elements are 0, then $\xi_{t2} \odot \tilde{\mathbf{A}}_t$ can be utilized for retaining the information in the high attention hierarchy, that is, long-term tendency. Third, $\xi_{t3} = \hat{\mathbf{A}}_t - \xi_{t1}$ denotes a vector whose elements in the range of $[0, \min(d_1, d_2)]$ are 1, and the other elements are 0, then $\xi_{t3} \odot \hat{\mathbf{A}}_t$ is used to update the information in the low attention hierarchy, that is, short-term tendency.

C. Proposed RUL Prediction Method for Servo Motor

With the acquired vibration signals of servo motors, the QFDCAE network is first employed to extract the motor HI vector that has the distinct performance degradation trend. Then the motor RUL is predicted by the developed GAHAU network based on the constructed HIs. The detailed prediction flowchart is shown in Fig. 5. To better comprehend the proposed RUL prediction methodology, its pseudocode is given in Table I.

- 1) *Vibration Signals Collection*: Under different working conditions, a series of life-cycle motor vibration signals are collected by accelerometers. The sampling time and the interval time are T_m and T_n , respectively. And the number of collected samples in a lifecycle is N .
- 2) *Motor HI Construction*: The cleaned vibration signals are transformed into frequency-domain data by fast Fourier transform (FFT), and then it is fed into QFDCAE for obtaining the HI vector $\mathbf{x} = [x_1, x_2, x_3, \dots, x_N]^T$. M HI values in $\mathbf{x}$ are selected to form the training samples $\mathbf{k} = [x_1, x_2, x_3, \dots, x_M]^T$.
- 3) *Motor RUL Estimation*: Based on the training vector $\mathbf{k}$, the Hankel matrix $\mathbf{V}$ is constructed. And the Hankel matrix $\mathbf{V}$ is written as

$$\mathbf{V} = \begin{bmatrix} x_1 & x_2 & \cdots & x_{M-i} \\ x_2 & x_3 & \cdots & x_{M-i+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i+1} & x_{i+2} & \cdots & x_M \end{bmatrix} = \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \\ \vdots \\ \mathbf{V}_{i+1} \end{bmatrix} \quad (16)$$

where i is the number of neurons in the input layer, and $\mathbf{V}_j = [x_j \ x_{j+1} \ \cdots \ x_{M-i+j-1}]$. The first i vectors of the matrix $\mathbf{V}$ are employed as the input of GAHAU, and the last vector $\mathbf{V}_{i+1}$ is used as the output in the training process of GAHAU. Evidently, the GAHAU network is equivalent to a mapping function f_m , and the loss function (15) can be rewritten as

$$\begin{aligned} \mathcal{L}(\mathbf{V}_{i+1}, f_m(\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_i)) \\ = \frac{1}{M-i} \sum_{n=1}^{M-i} [x_{i+n} - f_m(x_n, x_{n+1}, \dots, x_{n+i-1})]^2. \end{aligned} \quad (17)$$

To minimize the loss function $\mathcal{L}(\mathbf{V}_{i+1}, f_m(\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_i))$, the classical RNN learning algorithm, that is, real-time recurrent learning (RTRL), is utilized for training. After training the GAHAU network, the last i row vectors are used as input of trained GAHAU to calculate the output $\hat{\mathbf{V}}_{i+2}$ at the first prediction time (FPT), which is formulated as

$$\hat{\mathbf{V}}_{i+2} = f_m(\mathbf{V}_2, \mathbf{V}_3, \dots, \mathbf{V}_{i+1}). \quad (18)$$

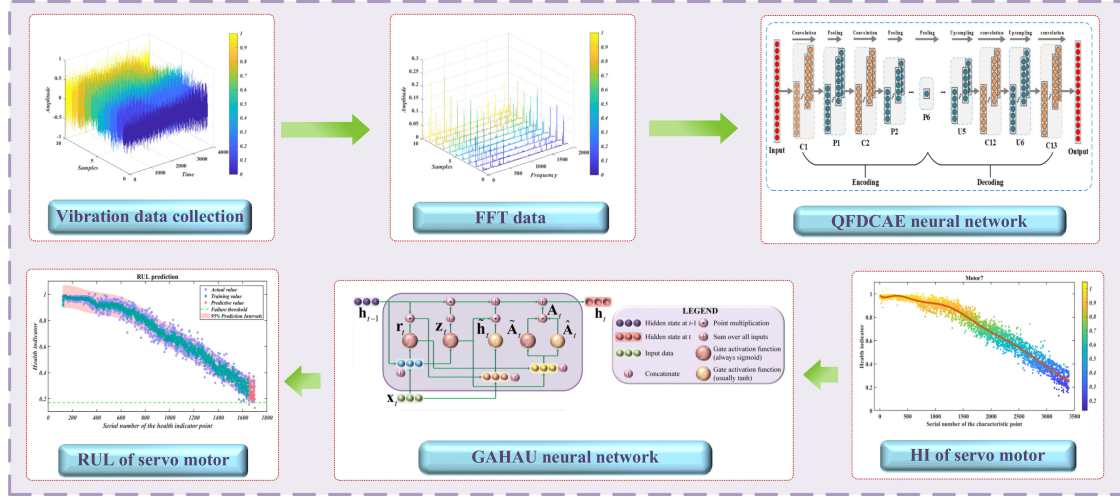


Fig. 5. Flowchart of the new RUL prediction methodology for servo motors.

TABLE I
PSEUDOCODE OF THE PROPOSED RUL PREDICTION METHODOLOGY

Algorithm : Proposed RUL estimation method for servo motor

Vibration signals collection:

Collect a series of life-cycle motor vibration signals.

Sampling time: T_m , interval time: T_n , the number of samples: N

Motor HI construction:

Clean vibration signals and transform them into frequency-domain signals.

Train QFDCAE neural network and construct servo motor HI vectors.

Motor RUL estimation:

Set the number of neurons in the input layer as i .

M HI values in the obtained HI vector are selected to form the training set.

Construct Hankel matrix $\mathbf{V}$:

$$\mathbf{V} = \begin{bmatrix} x_1 & x_2 & \cdots & x_{M-i} \\ x_2 & x_3 & \cdots & x_{M-i+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i+1} & x_{i+2} & \cdots & x_M \end{bmatrix} = \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \\ \vdots \\ \mathbf{V}_{i+1} \end{bmatrix}, \mathbf{V}_j = [x_j \ x_{j+1} \ \cdots \ x_{M-i+j-1}]$$

The first i vectors of $\mathbf{V}$ are utilized as input, the last vector $\mathbf{V}_{i+1}$ is employed as output.

For number of epochs, do

Train GAHAU network: $\min \mathcal{L}(\mathbf{V}_{i+1}, f_m(\mathbf{V}_1, \mathbf{V}_2, \cdots, \mathbf{V}_i))$

Optimization method: RTRL algorithm

End

If $\hat{y}_{M-i+j-1}(\hat{y}_{M-i+j-1} \in \hat{\mathbf{V}}_j)$ is not smaller than the preset failure threshold, **do**

Consecutively calculate output vectors, where the j th predicted output vector $\hat{\mathbf{V}}_j$ is denoted as:

$$\hat{\mathbf{V}}_j = f_m(\mathbf{V}_{j-i}, \mathbf{V}_{j-i+1}, \cdots, \mathbf{V}_{i+1}, \hat{\mathbf{V}}_{i+(j-i-1)})$$

End

Output: the servo motor RUL is calculated via $(j - M) * T_n$.

Then, the Hankel matrix $\mathbf{V}$ is renewed as

$$\mathbf{V} = (\mathbf{V}_1, \mathbf{V}_2, \cdots, \mathbf{V}_{i+1}, \hat{\mathbf{V}}_{i+2})^T. \quad (19)$$

By repeating the aforementioned process, the output vectors are predicted consecutively, where the j th predicted row vector $\hat{\mathbf{V}}_j$ is written as

$$\hat{\mathbf{V}}_j = f_m(\mathbf{V}_{j-i}, \mathbf{V}_{j-i+1}, \cdots, \mathbf{V}_{i+1}, \hat{\mathbf{V}}_{i+2}, \cdots, \hat{\mathbf{V}}_{i+(j-i-2)}, \hat{\mathbf{V}}_{i+(j-i-1)}). \quad (20)$$

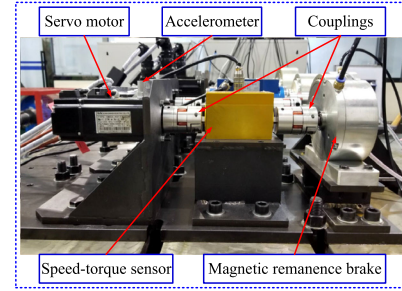


Fig. 6. Servo motor reliability test rig.

Finally, when the value of $\hat{y}_{M-i+j-1}(\hat{y}_{M-i+j-1} \in \hat{\mathbf{V}}_j)$ is smaller than the preset failure threshold, the servo motor RUL can be calculated via $(j - M) * T_n$. Furthermore, compared with the actual values, the performance of the proposed method is verified.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

For validating the predictive ability of GAHAU, multiple life-cycle tests of servo motors are performed, and the vibration signals are collected by vibration sensors for the RUL prediction. Then the RULs of servo motors are estimated via the proposed RUL prediction methodology. Moreover, to show the superiority of GAHAU, both the long-term and short-term predictive abilities of GAHAU are explored and compared to those of other existing prediction methods.

A. Experiment Description

The life-cycle experiment of the servo motor is conducted via a servo motor reliability test rig. As shown in Fig. 6, the test rig mainly consists of a testing servo motor, two couplings, a magnetic remanence brake, an accelerometer, and a speed-torque sensor. In this experiment, 9 servo motors were tested

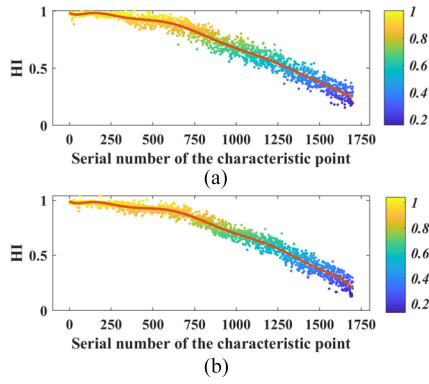


Fig. 7. Constructed HIs of motors 8 and 9. (a) Motor 8. (b) Motor 9.

under the half of rated torque and the half of rated speed. Servo motors 2, 5 and 9 were tested under the speed of 1500 r/min and the load of 1.2 Nm; servo motors 6 and 7 were tested under the speed of 1000 r/min and the load of 3 Nm; servo motor 1 was tested under the speed of 1500 r/min and the load of 0.64 Nm; servo motor 3 was tested under the speed of 1500 r/min and the load of 2.5 Nm; servo motor 4 was tested under the speed of 1000 r/min and the load of 2.1 Nm; servo motor 8 was tested under the speed of 1000 r/min and the load of 4.8 Nm. The life-cycle test was terminated when the motor degrade to a certain degree, however, there was no fault occurring at this time. In this work, when the fluctuation ratio of speed or torque is higher than a preset threshold (e.g., 2%), the test is stopped.

The accelerometer placed on the case of servo motor was used for collecting its vibration signals. The sampling frequency and sampling time span were separately set as 2000 Hz and 2 s, and the record interval between two adjacent samples was 30 min. The vibration datasets of motors 1–7 were chosen for training the HI construction model (i.e., QFDCAE), while the vibration datasets of motors 8 and 9 were used for validating the presented RUL prediction methodology.

B. RUL Prediction of Servo Motor

First, based on the acquired vibration signals of motors 8 and 9, their HIs are constructed by the trained QFDCAE. The obtained HI vectors of motors 8 and 9, respectively, have 3400 and 3398 elements. In order to reduce the training time of GAHAU, we perform the downsampling on the two HI vectors, then two tailored HI vectors are obtained, which include 1700 and 1699 points, respectively. The obtained HIs of motor 8 and 9 are, respectively, illustrated in Fig. 7(a) and (b). From Fig. 7, we can note that the motor HI has a decreasing tendency when the serial number of HI increases. Moreover, in early stage, the degradation rate of motor is small owing to the low deterioration velocity of a new motor. As time goes on, the degradation rate becomes larger than that in initial stage. Obviously, the two HI curves have obvious degradation trend, which basically accords with the performance degradation law of motor.

Second, the obtained HIs of testing motors are used as the input of GAHAU to predict motor RULs. For the HI vector

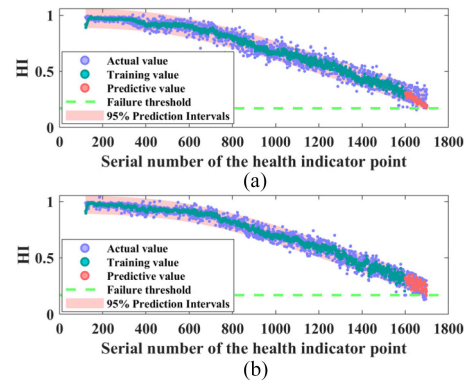


Fig. 8. Training curves, predicted curves, and actual curves for 100 predicted points. (a) Motor 8. (b) Motor 9.

of motor 8, the first 1600 points are applied to train GAHAU, then the last 100 points are predicted by the trained network. Similarly, for the HI vector of motor 9, the first 1599 points (i.e., the training set) are used to predict the rest 100 points. Obviously, the actual RULs of two testing motors are 100 h. Via calculating the mean value of the last five local minimum HI points of the testing motors, the failure threshold of RUL prediction is set as 0.17. Via a series of experiments, we design an optimized GAHAU network, and the number of neurons in input layer, hidden layer, and output layer are, respectively, chosen as 120, 21, and 1. Moreover, the learning rates of motors 8 and 9 are set as 0.01 and 0.024, respectively.

Finally, the RULs of motors 8 and 9 are predicted via the optimized GAHAU network. In order to make sure the prediction RUL is reliable and robust, the network with the same hyper-parameters is conducted on each motor HI for 10 times. Via calculating the mean value of 10 predicted RULs, the predictive RULs of motors 8 and 9 are obtained as 105.3 and 84.4 h, respectively. Considering the first test, the predicted HI curves, training HI curves, and 95% prediction intervals obtained by the MATLAB function “polyconf” of two motors are separately illustrated in Fig. 8(a) and (b). As shown in Fig. 8, most of the training and predicted HI values fall into the 95% predictive intervals. In particular, the degradation trends of the predicted and training HI curves are similar to that of the actual HI curve. Hence, the proposed approach is suitable for predicting the motor RUL because of the attention hierarchy in GAHAU.

C. Comparison With Other RUL Prediction Methods

For demonstrating the superiority of the proposed RUL prediction method over other RUL prediction methods, the classical prediction networks, such as GRU, LSTM, MMALSTM, are employed for comparison. For these RUL prediction approaches, the number of neurons in input layer, hidden layer, and output layer are, respectively, set to 120, 21, and 1. In addition, all the learning rates of the above networks are set to 0.02. Furthermore, the deep neural network (DNN) and polynomial segment fitting (PSF) method are also used for the RUL prediction. To quantitatively evaluate the predictive performance, the evaluation indexes that are defined in [27], such as mean absolute

TABLE II
EVALUATION INDEXES FOR DIFFERENT RUL PREDICTION METHODS

	MAE	NRMSE	MAPE	RMSE	Score
PSF	63.0000	1.7864	63.00%	66.0984	/
DNN	72.2000	1.0796	72.20%	85.3996	0.1521
GRU	48.5000	1.0118	48.50%	57.3672	0.2054
LSTM	38.6000	0.5371	38.60%	46.0804	0.2362
MMALSTM	34.1500	0.6414	34.15%	42.2345	0.4193
GAU	26.7000	0.3243	26.70%	33.1753	0.3677
GAHAU	15.4500	0.2619	15.45%	24.9730	0.4278

The bold entities highlight the best results obtained by GAHAU.

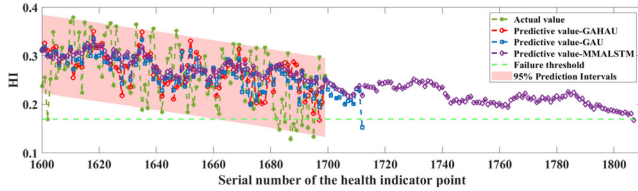


Fig. 9. Curves for predicting 100 points of motor 9 via GAHAU, GAU, and MMALSTM networks.

error (MAE), normalized root mean square error (NRMSE), mean absolute percentage error (MAPE), root mean square error (RMSE), and score, are utilized. Considering motors 8 and 9 simultaneously, the computed results of these evaluation indexes are shown in **Table II**.

From **Table I**, we can note the score of GAHAU is 0.4278 that is higher than those of other methods, and the prediction error of GAHAU is the lowest among these methods according to the results of MAE, NRMSE, MAPE, and RMSE. This is mainly because the proposed GAHAU network adequately considers the sequence information. We can also see that the scores of GAU and MMALSTM are much higher than that of DNN, GRU, and LSTM, and the prediction error is lower than PSF, DNN, GRU, and LSTM as well. This indicates that the attention mechanism is beneficial to improve the accuracy of motor RUL prediction. By dividing the sequence information of the attention gate in GAU, GAHAU gets the best predictive ability. From the above comparative results, we can conclude that the RUL prediction approach based on GAHAU has higher predictive performance than the current prediction methods.

In order to further demonstrate the predictive performance of MMALSTM, GAU, and GAHAU networks, the curves for predicting 100 HI points of motor 9 via GAHAU, GAU, and MMALSTM models are displayed in **Fig. 9**. From **Fig. 9**, it can be noted that the predicted HI points of GAHAU network are closer to the actual HI values than those of MMALSTM and GAU networks. In addition, the presented approach is more applicable for servo motor HI points prediction owing to the adaptive attention hierarchy mechanism in GAHAU. Via the constructed HI vector of motor 9, it is further proven that GAHAU has better fitting (predictive) ability than MMALSTM and GAU.

For evaluating the predictive performance of various approaches, the $\alpha - \lambda$ performance metric [4] is employed. α denotes the error limit with the actual RUL, and λ denotes the relative time distance between the estimated point and a motor's

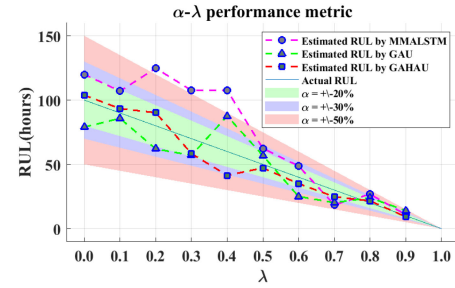


Fig. 10. Predictive performance analysis of various RUL prediction methods.

end of life (EOL). Taking $\alpha = 0.2$ and $\lambda = 0.5$ as an example, it measures whether the predicted RUL falls within 20% accuracy bound of the actual RUL at the halfway away from the EOL. We use the HI vectors of motor 8 with different known HI points to compare the uncertainty representation of the RUL prediction results obtained by GAHAU, GAU, and MMALSTM. Each method is performed 10 times, and the mean value of the predicted RULs is calculated. The uncertainty representations of these RUL prediction methods are shown in **Fig. 10**.

We can see from **Fig. 10** that the predicted results of the GAHAU will fall within 20% accuracy bound of the real RUL when the early degradation characteristics of motor appear (when $\lambda = 0.5$), and all predicted RUL results of the proposed method fall within 30% accuracy bound of the actual RUL. Moreover, the accuracy of RUL prediction becomes more accurate as the estimated HI point is approximate to the failure point. As the λ increases, the predictive performance of GAHAU increases dramatically. When λ is equal to 0.8, the prediction accuracy of GAHAU is 92%, while the prediction accuracy of MMALSTM and GAU are, respectively, 66% and 80%. Moreover, it can be observed from **Fig. 10** that all the predicted RULs of GAU fall within 50% accuracy bound of the actual RUL, and its prediction accuracy is superior to that of MMALSTM. From the above contrastive analysis, we can conclude that the proposed GAHAU has better performance on the uncertainty of RUL prediction than MMALSTM and GAU.

D. Experiment on Long-Term and Short-Term RUL Prediction

To further show the long-term and short-term predictive ability of GAHAU, it is used to predict the last 50 and 150 HI points of two motors, respectively. The predicted and training HI curves of motors 8 and 9 are displayed in **Figs. 11** and **12**, respectively.

From the above two figures, we can easily note that the predicted HI values are very close to the actual HI values, especially for 50 predicted points. Meanwhile, the degradation trends of the predictive and training HI vectors accord with that of the actual HI vectors. It then follows that the proposed RUL prediction approach has both long-term and short-term predictive ability owing to the adaptive attention hierarchy mechanism in GAHAU. Other RUL prediction approaches are also used for comparison. To quantitatively assess the long-term and short-term prediction performance, the evaluation indexes are calculated

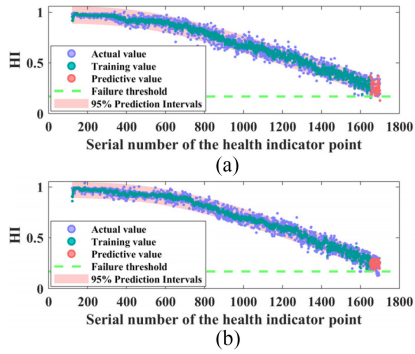


Fig. 11. Training and predicted HI vectors for 50 predicted points. (a) Motor 8. (b) Motor 9.

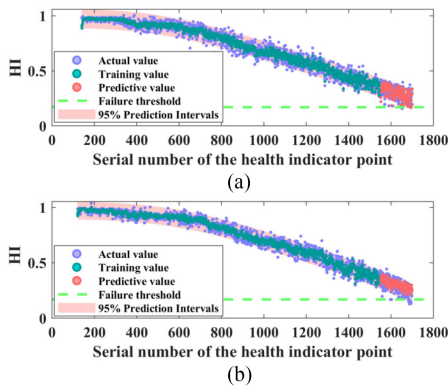


Fig. 12. Training and predicted HI curves for 150 predicted points. (a) Motor 8. (b) Motor 9.

TABLE III

EVALUATION INDEXES OBTAINED BY DIFFERENT PREDICTION METHODS FOR 50 PREDICTED POINTS

	MAE	NRMSE	MAPE	RMSE	Score
PSF	38.5000	3.4108	77.00%	39.2237	/
DNN	46.1000	0.8825	92.20%	50.8291	0.0436
GRU	29.8000	1.2299	59.60%	35.1753	0.2222
LSTM	39.1000	0.6130	78.20%	46.8914	0.1883
MMALSTM	22.7500	0.5230	45.50%	26.9583	0.2482
GAU	19.7000	0.5982	39.50%	24.3156	0.3329
GAHAU	7.8000	0.3080	15.60%	13.9212	0.5545

based on the predicting results of motors 8 and 9. Tables III and IV show the values of evaluation indexes obtained by various prediction methods. From Tables II and III, we can know that the score of GAHAU is much higher than those of other methods, and the MAE, NRMSE, MAPE, and RMSE of GAHAU are greatly lower than those of other methods. In addition, GAU and MMALSTM have better long-term and short-term predictive performance than PSF, DNN, GRU, and LSTM, since the used attention mechanism is able to process information with different weighting coefficients. As the sequence information plays a more role in time series prediction networks, GAHAU has strong predictive ability than GAU and MMALSTM, as listed in Tables III and IV. From these comparative results, we can conclude that the hierarchical attention information extremely helps to

TABLE IV

EVALUATION INDEXES OBTAINED BY DIFFERENT PREDICTION METHODS FOR 150 PREDICTED POINTS

	MAE	NRMSE	MAPE	RMSE	Score
PSF	76.0000	1.2294	50.67%	90.9725	/
DNN	97.2500	1.0513	64.83%	109.5936	0.1478
GRU	78.1500	0.9009	52.10%	85.9014	0.1566
LSTM	76.8000	0.3915	51.20%	88.4313	0.1241
MMALSTM	45.6000	0.4607	30.40%	65.5057	0.3758
GAU	62.9000	0.3774	41.93%	78.9607	0.1470
GAHAU	22.0000	0.1850	14.67%	28.5797	0.4231

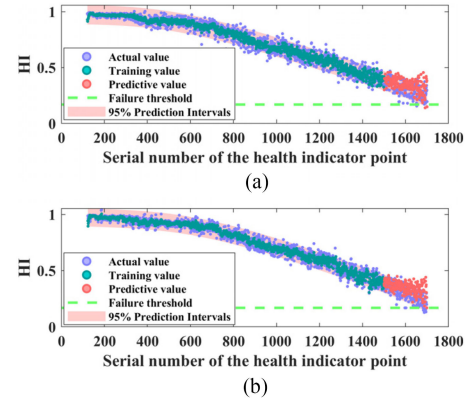


Fig. 13. Training and predicted HI curves for 200 predicted points. (a) Motor 8. (b) Motor 9.

TABLE V

EVALUATION INDEXES OBTAINED BY DIFFERENT PREDICTION METHODS FOR 200 PREDICTED POINTS

	MAE	NRMSE	MAPE	RMSE	Score
PSF	94.5000	0.8967	47.25%	94.6071	/
DNN	122.2000	0.5991	61.10%	131.3750	0.0988
GRU	91.9000	0.7412	45.95%	99.0994	0.2358
LSTM	96.4500	0.5122	48.23%	117.5651	0.2453
MMALSTM	88.9000	0.3709	44.45%	98.5581	0.1308
GAU	84.0000	0.3795	42.00%	95.8953	0.1242
GAHAU	62.7500	0.3667	31.38%	87.3725	0.3419

improve the prediction accuracy and robustness, especially for the long-term RUL prediction.

Next, in order to further demonstrate the long-term predictive capacity of GAHAU, the HI vectors of motors 8 and 9 are employed to predict the last 200 HI points, i.e., 1500 and 1499 HI points are known. The training and predictive HI curves for 200 predicted points are shown in Fig. 13. For motor 8, the MAE is calculated as 75.8, and the prediction accuracy which is defined as $1 - \text{MAPE}$ is calculated as 62.1%. For motor 9, the MAE is computed as 49.7, and the prediction accuracy is computed as 75.1%. Based on the RUL prediction results of motors 8 and 9, the relevant indexes for quantitatively evaluating the predictive performance of the proposed approach are computed. The evaluation indexes acquired by different RUL prediction methods are shown in Table V. It can see from Table V that the score of GAHAU is still much higher than those of other RUL estimation methods, while the MAE, NRMSE, MAPE, and RMSE of the proposed method are still greatly lower

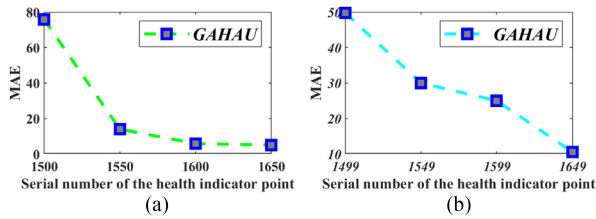


Fig. 14. MAE values for different predicted HI points. (a) Motor 8. (b) Motor 9.

than those of other existing RUL prediction approaches. Moreover, the predicted performance of MMALSTM and GAU is better than PSF, DNN, GRU, and LSTM. This is mainly because that the attention mechanism in these networks can address information differently. The above experimental results reveal that the proposed GAHAU still has certain long-term predictive ability, and the hierarchical attention information can be helpful for enhancing the long-term RUL predicted performance.

At last, the MAE values for 50, 100, 150, and 200 predictive points are calculated to comprehensively evaluate the predictive ability of GAHAU. The MAE values of motors 8 and 9 for different predicted points are separately illustrated in Fig. 14(a) and (b). We can note from this figure that MAE decreases with the increase of the known HI points, that is, the prediction error becomes smaller when more known HI values are used for training. This is mainly because that more intrinsic degradation information will be mined from the more known HI points by GAHAU.

V. CONCLUSION

In this article, a novel RUL prediction methodology for servo motors based on the principle of time series forecasting was developed. An adaptive hierarchical attention unit was designed to automatically divide the sequence information of attention gate into three attention hierarchies. Then GAHAU neural network was constructed, and its updating formulas were derived. With the servo motor HI constructed by the QFDCAE network and vibration signals, GAHAU could implement the RUL prediction of servo motors. As the attention hierarchy mechanism could adaptively mine the sequence information of attention gate, the GAHAU network possessed strong predictive ability. Via the life-cycle experiments of servo motors, the effectiveness and superiority of the proposed methodology was validated. Compared to PSF, DNN, GRU, LSTM, MMALSTM, and GAU, GAHAU has higher prediction accuracy and robustness in both long-term and short-term prediction. In future work, a better hierarchy mechanism can be explored to improve the performance of long-term RUL prediction.

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